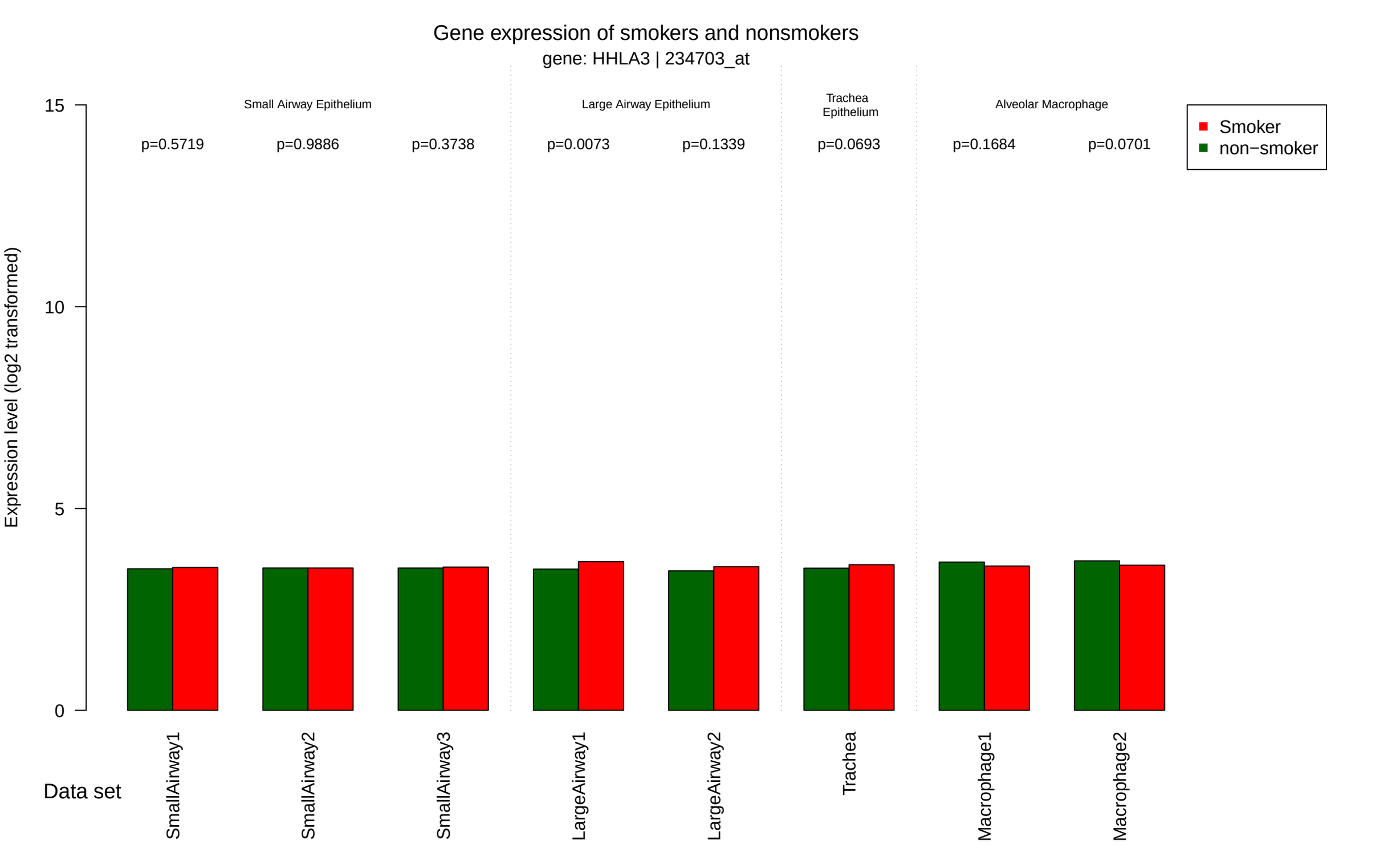
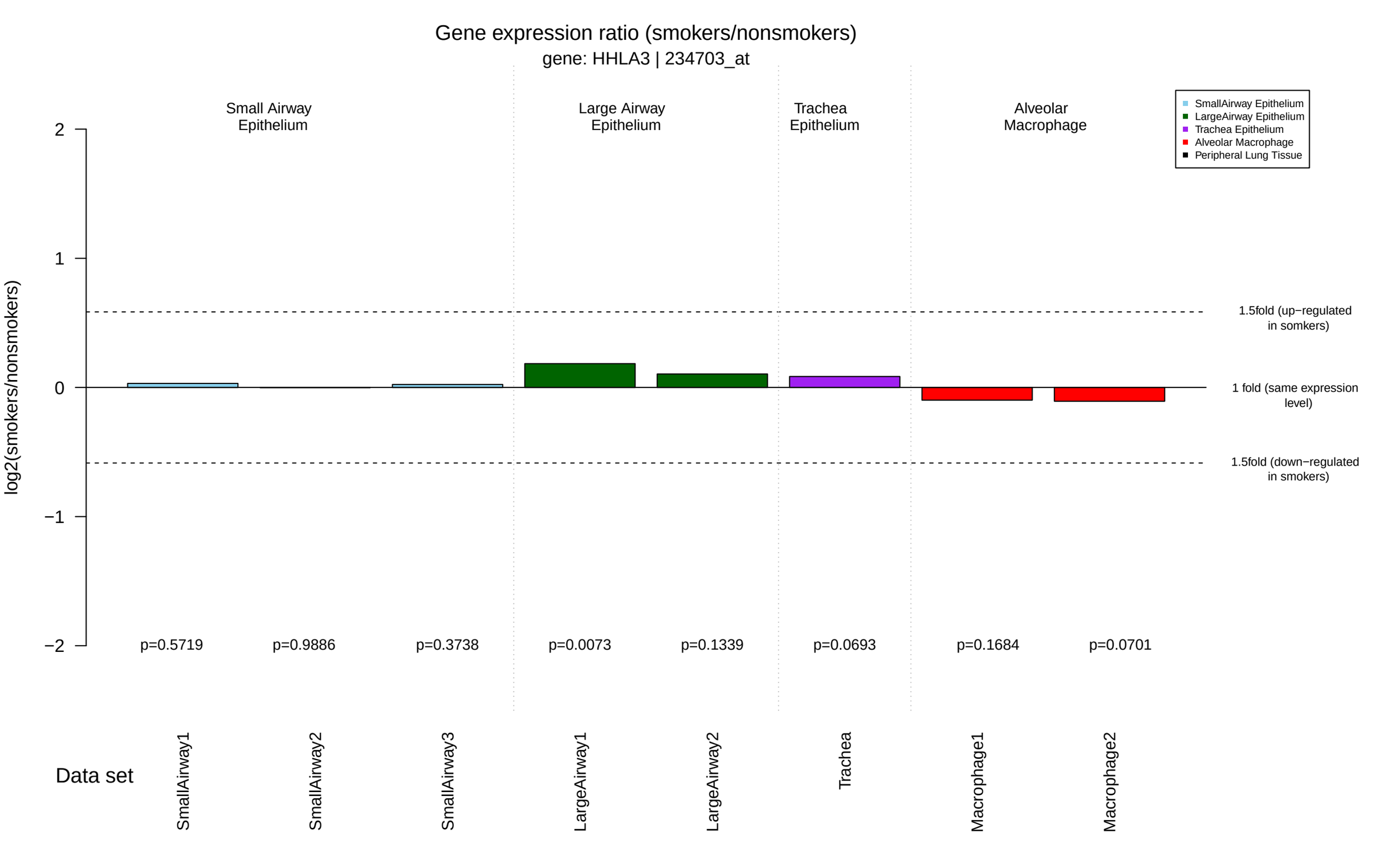
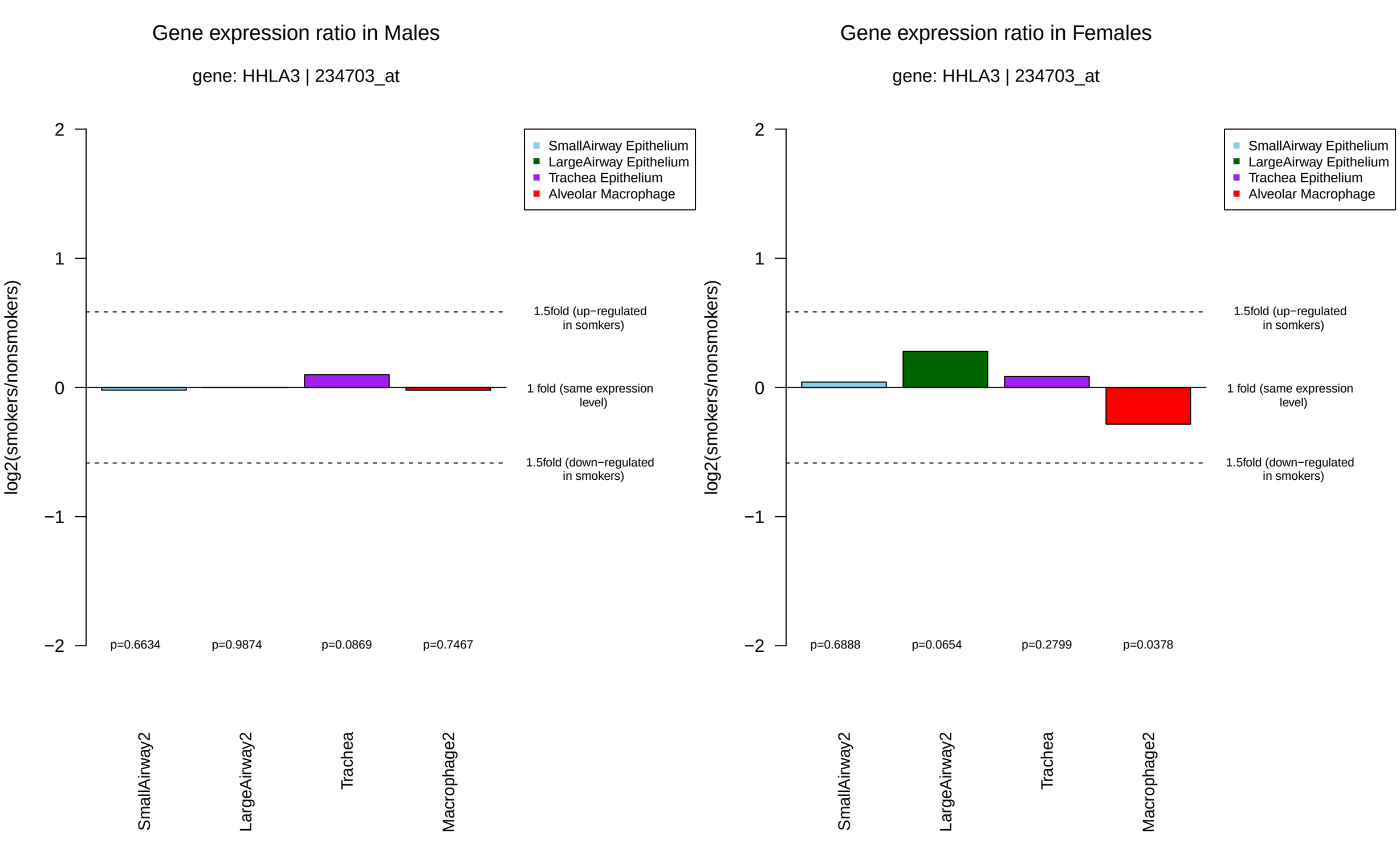
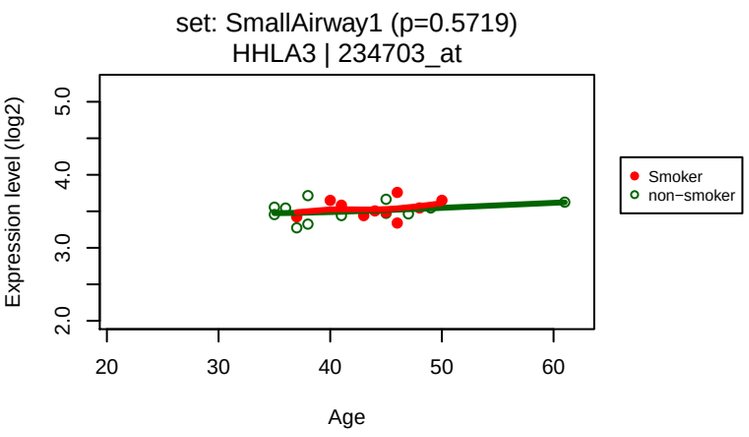
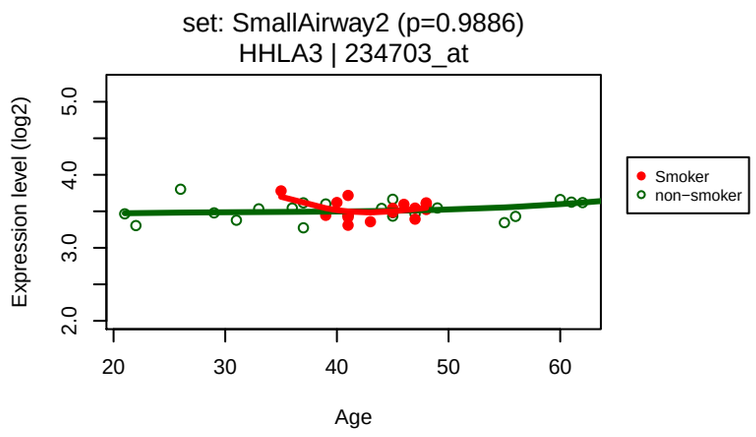
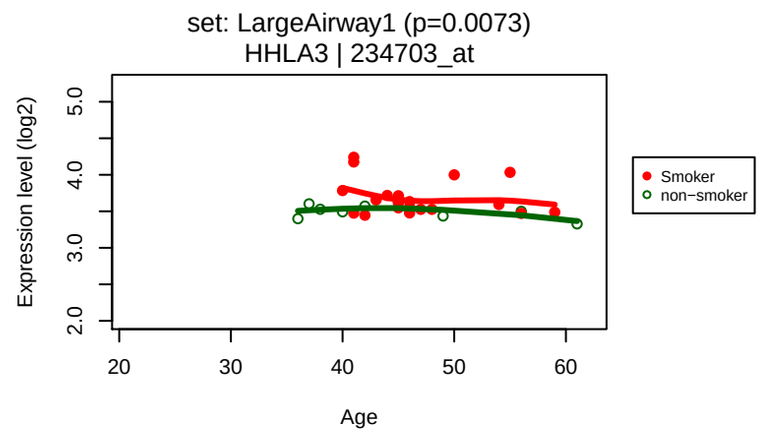

Gene expression of smokers and nonsmokers
gene: HHLA3 | 234703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5719
p=0.9886
p=0.3738
p=0.0073
p=0.1339
p=0.0693
p=0.1684
p=0.0701
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

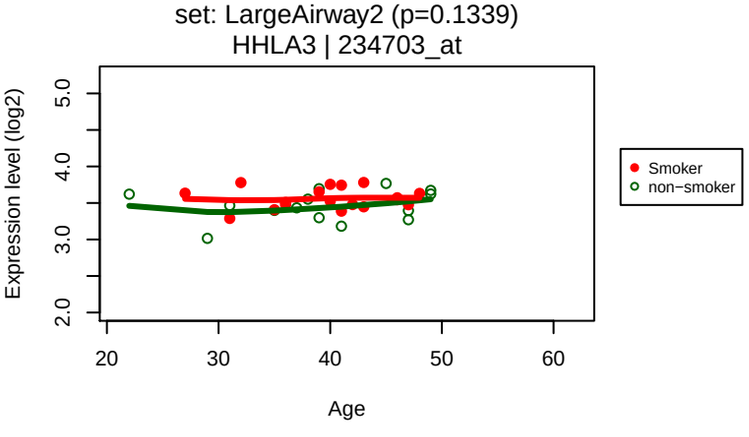

Gene expression ratio (smokers/nonsmokers)
gene: HHLA3 | 234703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5719
p=0.9886
p=0.3738
p=0.0073
p=0.1339
p=0.0693
p=0.1684
p=0.0701
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

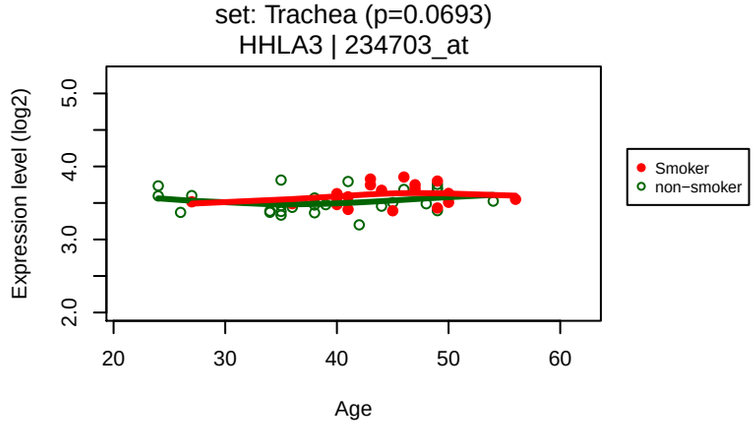

Gene expression ratio in Males
gene: HHLA3 | 234703_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6634
p=0.9874
p=0.0869
p=0.7467
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HHLA3 | 234703_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6888
p=0.0654
p=0.2799
p=0.0378
SmallAirway2
LargeAirway2
Trachea
Macrophage2

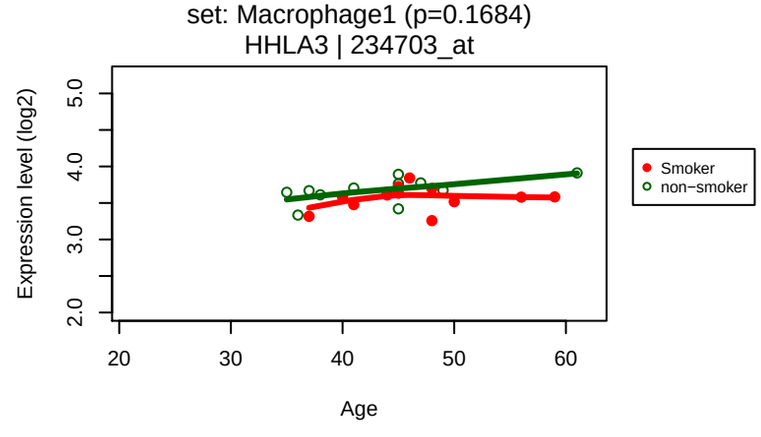

set: SmallAirway1 (p=0.5719)
HHLA3 | 234703_at
Expression level (log2)
Age
Smoker
non−smoker

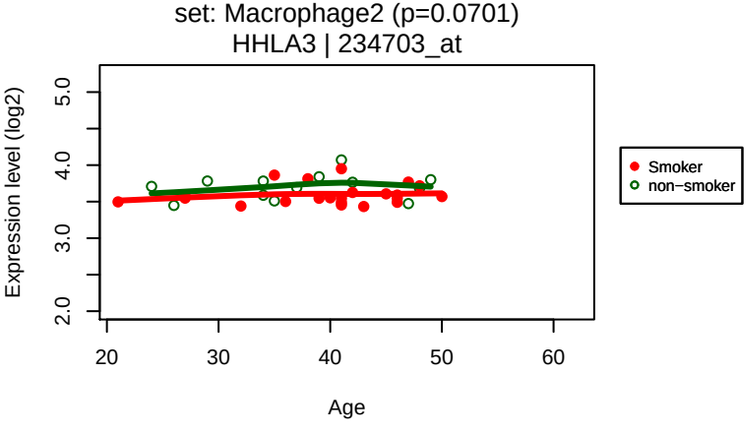

set: SmallAirway2 (p=0.9886)
HHLA3 | 234703_at
Expression level (log2)
Age
Smoker
non−smoker

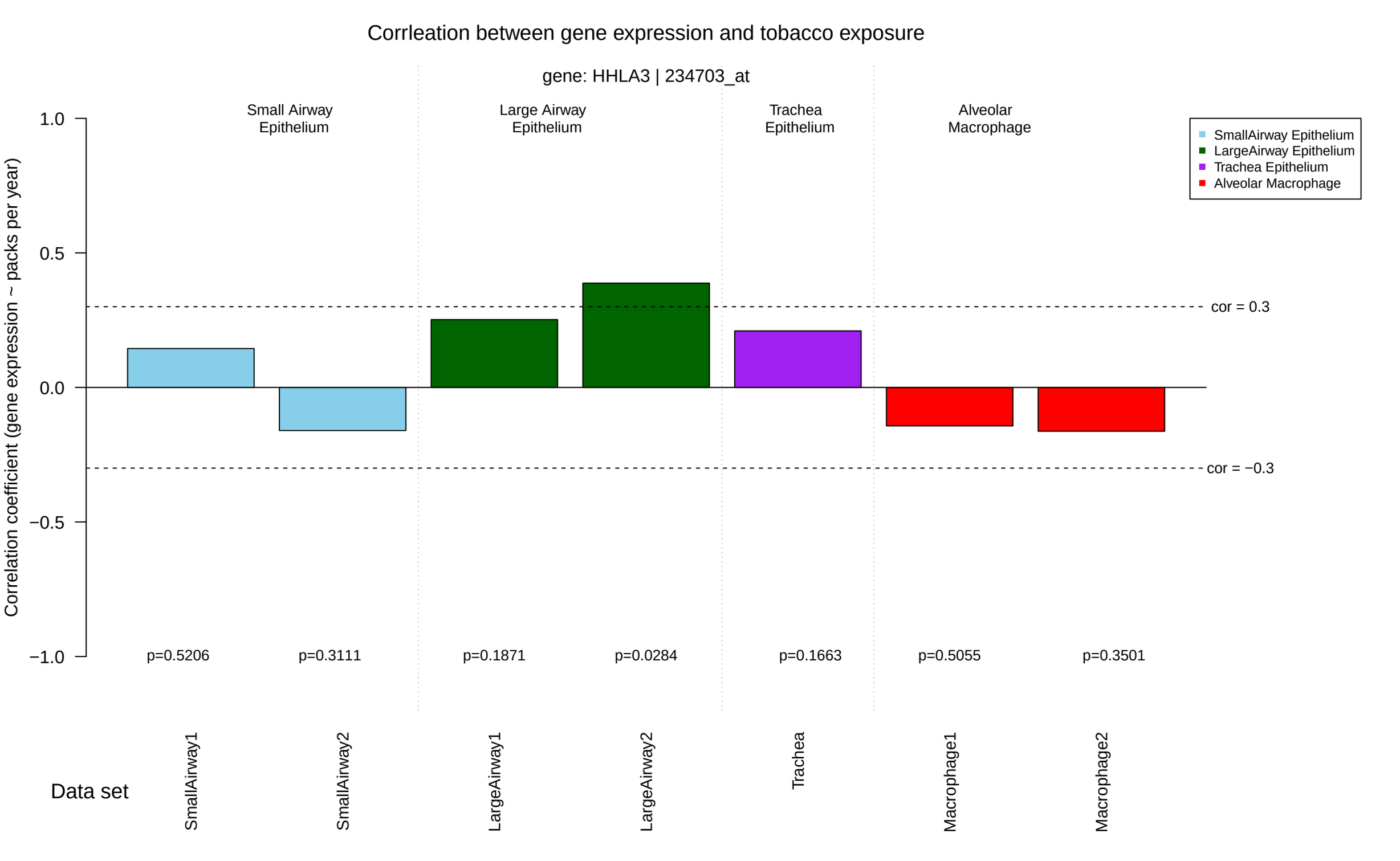

set: LargeAirway1 (p=0.0073)
HHLA3 | 234703_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1339)
HHLA3 | 234703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0693)
HHLA3 | 234703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1684)
HHLA3 | 234703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0701)
HHLA3 | 234703_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HHLA3 | 234703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5206
p=0.3111
p=0.1871
p=0.0284
p=0.1663
p=0.5055
p=0.3501
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2